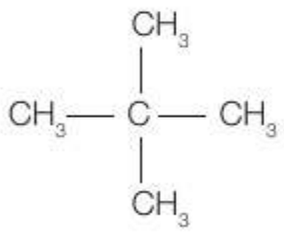
4 EXAM PRACTICE

1 An organic compound is shown by this formula.



What type of formula is this?

A displayed formula

B general formula

C molecular formula

D structural formula [1]

(Total for Question 1 = 1 mark)

2 Which statement is correct for the members of a homologous series?

A They have similar boiling points

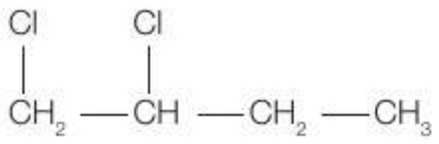
B Their molecular formulae differ by CH_3

C They contain the same functional group

D Their chemical properties are different [1]

(Total for Question 2 = 1 mark)

3 What is the IUPAC name for this compound?



A 3,4-dichlorobutene

B 1,2-dichlorobutane

C 3,4-dichlorobutane

D 1,2-dichlorobutene [1]

(Total for Question 3 = 1 mark)

4 Which fuel can be carbon neutral?

A ethanol

B natural gas

C petrol

D wood [1]

(Total for Question 4 = 1 mark)

5 Which equation represents a substitution reaction?

A $C_2H_4 + H_2O \rightarrow C_2H_5OH$

B $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$

C $C_2H_6 + F_2 \rightarrow C_2H_5F + HF$

D $C_2H_6 \rightarrow C_2H_4 + H_2$ [1]

(Total for Question 5 = 1 mark)

6 Which equation represents a termination step in the reaction between chlorine and chloromethane?

A $\bullet CH_2Cl + Cl\bullet \rightarrow CH_2Cl_2$

B $H\bullet + Cl\bullet \rightarrow HCl$

C $\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet$

D $CH_3Cl + Cl_2 \rightarrow CH_2Cl_2 + HCl$ [1]

(Total for Question 6 = 1 mark)

7 The table lists the boiling temperatures of some alkanes.

Alkane	Molecular formula	Boiling temperature / K
butane	C_4H_{10}	273
pentane	C_5H_{12}	309
hexane	C_6H_{14}	342
heptane	C_7H_{16}	372
octane		399
nonane	C_9H_{20}	
decane	$C_{10}H_{22}$	447

(a) Give the molecular formula of octane. [1]

(b) (i) Explain the trend in boiling temperature of the alkanes. [2]

(ii) Predict a value for the boiling temperature of nonane. [1]

(c) Long chain alkanes, such as decane, can be cracked into shorter chain alkanes and alkenes.

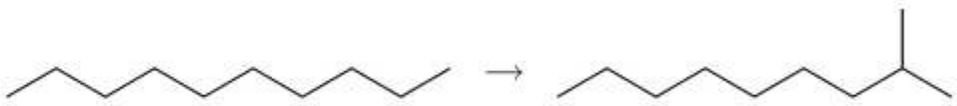
(i) Write an equation for the cracking of decane into octane and ethene. [1]

(ii) The ethene produced can be converted into ethanol by direct hydration with steam.

Write an equation for this reaction and state the conditions that are used in industry. [4]

(d) Reforming is a process used in the production of petrol. Unbranched-chain alkanes can be reformed to produce either branched-chain alkanes or cycloalkanes.

The equation shows the reforming of decane into 2-methylnonane.



(i) Using skeletal formulae, write an equation for the reforming of decane into 2,3-dimethyloctane. [1]

(ii) Using skeletal formulae, write an equation for the reforming of heptane into methylcyclohexane. [2]

(iii) State why reforming is used in the production of petrol. [1]

(Total for Question 7 = 13 marks)

8 Compound **Y** is a hydrocarbon containing 85.7% of carbon by mass.

(a) (i) Calculate the empirical formula of **Y**. [2]

(ii) The molar mass of **Y** is 56 g mol^{-1}. Show that the molecular formula of **Y** is C_4H_8. [1]

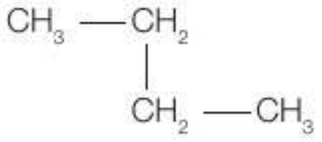
(b) There are six isomers for compound **Y**; four unsaturated molecules and two saturated molecules.

Draw a displayed formula for each of the five isomers and name each compound. [6]

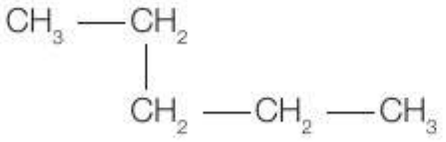
(Total for Question 8 = 9 marks)

9 The structural formulae of four hydrocarbons, **A**, **B**, **C** and **D**, are shown.

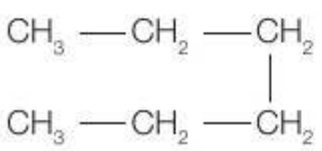
A



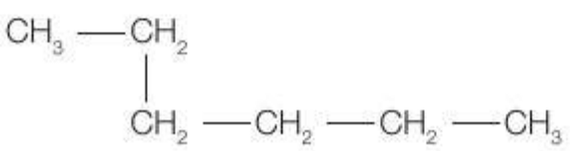
B



C



D



(a) Identify the homologous series to which all these hydrocarbons belong.

Give a reason for your answer. [2]

(b) Explain which structural formulae represent only one compound. [2]

(c) Give the structural formulae for the two isomers of **B**.

Give the IUPAC name for each isomer. [4]

(Total for Question 9 = 8 marks)